

Ultra-short PPG windows



Phase-space reconstruction



Complementary nonlinear characterization



Prediction

Finite-horizon
forecastability

RQA

Recurrence
organization

LLE

Local trajectory
divergence



PPS surrogate testing

Organization beyond tested
noisy pseudoperiodic null (RQ1)

Awake-Drowsy comparison

State-related dynamical
reorganization (RQ2)